Tower Performance Report

Cooling Tower P2 CT1
Owner: GOOD PARTNERS
Project: Cooling Tower P2
Location: Nan-Tze Kaohsiung
Manufacturer: SPX MARLEY
Tower Type: Induced Draft

Cooling Tower Design and Test Data

Parameters	**Design**	**Test**
Water Flow Rate	201.3 l/s	202.8 l/s
Hot Water Temp.	37.00 ℃	34.80 ℃
Cold Water Temp.	32.00 ℃	29.60 ℃
Wet Bulb Temp.	29.00 ℃	25.60 ℃
Dry Bulb Temp.	38.46 ℃	28.50 ℃
Fan Driver Power	37.00 kW	32.24 kW
Barometric Pressure	101.000 kPa	101.000 kPa
Liquid to Gas Ratio	1.400	1.461

Cold Water Temperatures vs. Range

At25.60 ℃ Test Wet Bulb

Range	**181.2 l/s**	**201.3 l/s**	**221.5 l/s**
3.89 ℃	28.07 ℃	28.47 ℃	29.16 ℃
5.00 ℃	28.68 ℃	29.28 ℃	30.00 ℃
6.11 ℃	29.19 ℃	29.94 ℃	30.80 ℃

Cold Water Temperature vs. Water Flow

At25.60 ℃ Test Wet Bulb and 5.20 ℃ Test Range

181.2 l/s	**201.3 l/s**	**221.5 l/s**
28.78 ℃	29.41 ℃	30.14 ℃

Exit Air Properties

	Wet Bulb Temp	**Density**	**Sp. Vol.**	**Enthalpy**
Design	34.10	1.12208	0.9224	123.9667
Test	31.95	1.13268	0.9101	110.9293

Test Results

Adjusted Flow	**Predicted Flow**	**CWT Deviation**	**Tower Capability**
213.0 l/s	206.9 l/s	-0.15 ℃	103.0%

This test result is only certified by CTI if the test data was collected by a CTI Licensed Testing Agency. See www.cti.org for an agency list.

Produced by CTI Toolkit 3.2

冷卻水塔性能量值記錄表

設備編號: CT1

入口濕球溫度
每次測10秒,1小時測6次.

時間	溫度(^{0}C)
13:30	25.8
13:40	25.6
13:50	25.5
14:00	25.5
14:10	25.6
14:20	25.6
平均濕球溫度 **A**	25.60

環境風速

時間	風速(m/s)
13:30	2.3
13:40	1.9
13:50	2.5
14:00	2.7
14:10	2.2
14:20	2.4
平均風速	2.3

冷水溫度

時間	溫度(^{0}C)
13:30	29.5
13:40	29.7
13:50	29.7
14:00	29.6
14:10	29.5
14:20	29.6
平均溫度 **B**	29.60

熱水溫度

時間	溫度(^{0}C)
13:30	34.8
13:40	34.8
13:50	34.7
14:00	34.8
14:10	34.9
14:20	34.9
平均溫度 **C**	34.80

循環水量

時間	流量 (CMH)
13:30	731
13:40	733
13:50	729
14:00	721
14:10	733
14:20	734
平均流量	730.10

馬達電流值 <u>50.7A</u> **電壓：<u>432V</u>**

時間	日期	單位	瞬間量
13:31	2013/5/17	m3/h	731.2
13:32	2013/5/17	m3/h	731.4
13:33	2013/5/17	m3/h	729.3
13:34	2013/5/17	m3/h	730.4
13:35	2013/5/17	m3/h	722.2
13:36	2013/5/17	m3/h	719.7
13:37	2013/5/17	m3/h	723.5
13:38	2013/5/17	m3/h	732.1
13:39	2013/5/17	m3/h	737.7
13:40	2013/5/17	m3/h	733.2
13:41	2013/5/17	m3/h	735.6
13:42	2013/5/17	m3/h	736.7
13:43	2013/5/17	m3/h	731.1
13:44	2013/5/17	m3/h	739.9
13:45	2013/5/17	m3/h	718.2
13:46	2013/5/17	m3/h	723.3
13:47	2013/5/17	m3/h	724.1
13:48	2013/5/17	m3/h	722.8
13:49	2013/5/17	m3/h	717.1
13:50	2013/5/17	m3/h	729.2
13:51	2013/5/17	m3/h	734.3
13:52	2013/5/17	m3/h	733.1
13:53	2013/5/17	m3/h	724.4
13:54	2013/5/17	m3/h	725.8
13:55	2013/5/17	m3/h	722.8
13:56	2013/5/17	m3/h	720.6
13:57	2013/5/17	m3/h	718.3
13:58	2013/5/17	m3/h	719.5
13:59	2013/5/17	m3/h	721.1
14:00	2013/5/17	m3/h	724.2
14:01	2013/5/17	m3/h	726.5
14:02	2013/5/17	m3/h	727.8
14:03	2013/5/17	m3/h	726.9
14:04	2013/5/17	m3/h	721.4
14:05	2013/5/17	m3/h	728.5
14:06	2013/5/17	m3/h	725.4
14:07	2013/5/17	m3/h	732.8
14:08	2013/5/17	m3/h	731.6
14:09	2013/5/17	m3/h	725.7
14:10	2013/5/17	m3/h	731.1
14:11	2013/5/17	m3/h	736.3
14:12	2013/5/17	m3/h	731.6
14:13	2013/5/17	m3/h	739.8
14:14	2013/5/17	m3/h	720.5
14:15	2013/5/17	m3/h	718.5
14:16	2013/5/17	m3/h	719.7
14:17	2013/5/17	m3/h	731.3
14:18	2013/5/17	m3/h	724.4
14:19	2013/5/17	m3/h	733.1
14:20	2013/5/17	m3/h	733.6
14:21	2013/5/17	m3/h	730.7
14:22	2013/5/17	m3/h	721.1
14:23	2013/5/17	m3/h	718.5
14:24	2013/5/17	m3/h	739.7
14:25	2013/5/17	m3/h	722.4
14:26	2013/5/17	m3/h	724.1
14:27	2013/5/17	m3/h	725.8
14:28	2013/5/17	m3/h	723.3
14:29	2013/5/17	m3/h	720.1
14:30	2013/5/17	m3/h	719.6

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